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Course Program:- Btech CSE SY

Subject:- DLM

DLM lab practical. 8

* Aim:- Demux using 74139 for code converter.

* Theory:-

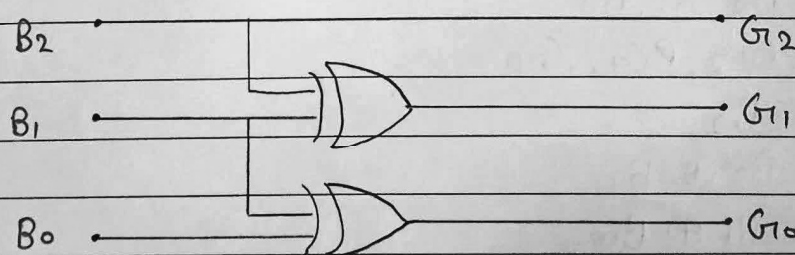
Binary codes: It is a weighted code i.e, it is a code in which weight is assigned to every symbol position in the code. The positional weights in binary code are shown below

Binary $\rightarrow$ --- 2^4 2^3 2^2 2^1 2^0 2^{-1} 2^{-2} ---

Decimal $\rightarrow$ --- 16 8 4 2 1 $1/2$ $1/4$ ---

Gray codes:- Non-weighted code i.e, it does not have any specific or fixed weighted to each symbol position in the code word. The unique feature of gray code is that at a time only one bit change in other word in gray code, every new code differs from previous in terms of one single bit.

Diagram:-





Truth Table:-

Decimal	Binary			Gray		
	B ₂	B ₁	B ₀	G ₂	G ₁	G ₀
0	0	0	0	0	0	0
1	0	0	1	0	0	1
2	0	1	0	0	1	1
3	0	1	1	0	1	0
4	1	0	0	1	1	0
5	1	0	1	1	1	1
6	1	1	0	1	0	1
7	1	1	1	1	0	0

from truth table it is observed that the o/p G₂ is high for minterms M₄, M₅, M₆ and M₇ hence equation defining G₂ o/p can be written as $G_2 = \sum m(4, 5, 6, 7)$
The o/p G₁ is high for minterms M₂ and M₃, M₄, M₅
Hence equation defining. The o/p G₀ is high for minterms M₁, M₂, M₅, M₆. Hence equation defining G₁ o/p can be written as $G_0 = \sum m(1, 2, 5, 6)$. The boolean expression can be implemented using IC 74153 as 3:8 decoder by using connecting its relevant o/p to 4-input NAND gates.

* formulae:

$$I/P = B_2, B_1, B_0$$

$$o/p = G_2, G_1, G_0$$

$$G_2 = B_2$$

$$G_1 = B_2 \oplus B_1$$

$$G_0 = B_1 \oplus B_0$$



* Procedure:

- Switch on the circuit, by pressing main switch.
- Select appropriate I/p
- After giving I/p, observe corresponding o/p.
- Verify the result.

* conclusions:-

I learned about using 74139 decoder for code converters helped me understand how decoders can facilitate the translation between different codes. The 74139 and a dual 2:4 line decoder is used to decode inputs into specific o/p lines in converting between various binary codes.